

Teo extension entera ideal primo maximal iff maximal

Teorema 1. *Sea $A \subset B$ una extensión entera de anillos y sea $\mathfrak{p} \subset B$ un ideal primo. Entonces, $\mathfrak{p}$ es maximal $\iff \mathfrak{p} \cap A$ es maximal.*

Demostración: Por el ejer-extension-entera-cociente/??, $A/\mathfrak{p} \cap A \subset B/\mathfrak{p}$ es una extensión entera de anillos. Además, como $\mathfrak{p}$ es primo, $\mathfrak{p} \cap A$ es primo, luego por la prop-ideal-primo-iff-cociente-di-integridad/??, $B/\mathfrak{p}$ y $A/\mathfrak{p} \cap A$ son dominios de integridad.

Entonces, por el teo-extension-entera-dominios-integridad-imp-cuerpo-iff-cuerpo/??, $A/\mathfrak{p} \cap A$ es un cuerpo $\iff B/\mathfrak{p}$ es un cuerpo. Luego, por la prop-ideal-maximal-iff-cociente-cuerpo/?, $\mathfrak{p} \cap A$ es maximal $\iff \mathfrak{p}$ es maximal. ■

Referenciado en

- teo-extension-entera-ideal-primo-imp-exists-ideal-primo-contrae

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